Summary

Most prokaryotes have one of three general shapes coccus (round), bacillus (rod), or spiral, based on the configuration of the cell wall. Two types of spiral cells are spirochetes and spirlla. Shape and arrangement of cells are key factors for describing prokaryotes. Arrangements of cells are based on the number of planes in which a given species divides.

Cocci can divide in many planes to form pairs, chains, packets, or clusters. Bacilli divide only in the transverse plane. If they remain attached, they form chains or palisades.

Some bacterial cell walls are covered by capsules or slime which protect the cell from phagocytosis, drying and nutrient loss. Fimbriae and Pili are involved in attachment and in transfer of genetic information between bacterial cells. Flagella are involved in cell motility.

The cell envelope is the complex boundary structure surrounding a bacterial cell. In Gram negative bacteria, the envelope consists of outer membrane, the cell wall, and the cell membrane. Gram positive bacteria have only cell wall and cell membrane. Gram positive bacteria have thick murein and teichoic acid. The cell walls of Gram negative bacteria are thinner and have wide periplasmic space. The outer membrane of Gram negative cells contains LPS toxic to mammalian hosts. The cell membrane is typically composed of phospholipids and proteins, and it performs many metabolic functions as well as transport activities.

The cytoplasm of bacterial cell serve as a solvent for material used in all functions. The genetic material of bacteria is DNA and the genes are arranged on larger, circular chromosomes. Additional genes are carried on plasmid. Bacterial ribosomes are dispersed is the cytoplasm in chains and are also embedded in the cell membrane.

Bacteria may store nutrients in their cytoplasm, in form of inclusions. Inclusion vary in structure and the materials that are stored. A few bacteria produce dormant bodies called endospores, which are the hardiest of all life forms, survival for hundreds or thousands of years. The genus *Bacillus* and *Clostridium* are spore forming deadly pathogens.

Eukaryotes are cells with nucleus and membrane bound organelles. Cell structures common to most eukaryotes are the cell membrane, nucleus, vacuoles, mitochondria, endoplasmic reticulum, golgi apparatus and a cytoskeleton. Cell wall, chloroplast and locomotory organelles are present in some eukaryote groups.

Evaluation

Multiple choice questions

1. The arrangement of flagella on cell surface can sometimes help in the identification of an organism for example, *Escherichia coli* to has flagella throughout the cell surface that is referred to as.
   a. Lophotrichous
   b. Monotrichous
   c. Peritrichous
   d. None of the above

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